

JomCode Coding Bootcamp

Dictionaries, sets and records

Understand, predict, type, run, then practice.

What you will be able to do

- 01 Model records using dictionaries and choose safe lookup behavior.
- 02 Use sets for uniqueness and membership.
- 03 Transform lists of records without conflating sequence order and identity.

Plain meaning → technical view → use

PLAIN MEANING

Look up a value by name or keep only unique values.

TECHNICAL VIEW

Dictionaries map keys to values; sets support unique membership checks.

WHEN YOU USE IT

Represent a workshop record and detect duplicate participant IDs.

TIP

Choose a dictionary for one record and a set for uniqueness.

Worked example 1: Record lookup



chapter_08_dictionaries_sets_and_records.py

```
1 workshop = {"id": "w1", "title": "Python", "capacity": 20}
2 print(workshop["title"])
3 print(workshop.get("venue", "TBC"))
4 print("title" in workshop)
```

PREDICT When should a missing key be an error?

Example 1: output and reason

TERMINAL >_ EXPECTED OUTPUT

```
Python  
TBC  
True
```

WHY THIS HAPPENS

Required fields should fail visibly. Optional presentation fields may use an explicit default.

Exercise 1: Default venue

BUILD

Read an optional venue and use TBC only when the key is absent.

CHECK

Absent => TBC. Present empty string remains empty and needs later validation.

IF STUCK

A default for absence differs from a fallback for false-like values.

Worked example 2: Duplicate identifiers



chapter_08_dictionaries_sets_and_records.py

```
1 participant_ids = ["u2", "u1", "u2"]
2 unique_ids = set(participant_ids)
3 print(sorted(unique_ids))
4 print("u1" in unique_ids)
5 print(unique_ids & {"u1", "u3"})
```

PREDICT Why sort before presenting the result?

Example 2: output and reason

TERMINAL >_ EXPECTED OUTPUT

```
['u1', 'u2']  
True  
{'u1'}
```

WHY THIS HAPPENS

A set represents uniqueness, not a stable display sequence.

Exercise 2: Registration uniqueness

BUILD

Create a sorted unique list from ["p3", "p1", "p3", "p2"].

CHECK

=> [p1,p2,p3]. Empty input => [].

IF STUCK

Use set for identity, sorted for stable output.

Worked example 3: Filter records



chapter_08_dictionaries_sets_and_records.py

```
1 workshops = [  
2     {"id": "w1", "capacity": 0},  
3     {"id": "w2", "capacity": 5},  
4 ]  
5 open_ids = [row["id"] for row in workshops if row["capacity"] > 0]  
6 print(open_ids)
```

PREDICT Is w2 the same concept as list index 1?

Example 3: output and reason

TERMINAL	>_	EXPECTED OUTPUT
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<pre>['w2']</pre>		
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WHY THIS HAPPENS

A stable ID survives sorting and filtering. The list index changes with presentation order.

Exercise 3: Aggregate capacities

BUILD

Calculate the total capacity from two workshop dictionaries, including an empty-list case.

CHECK

Capacities 3 and 7 => 10. Empty => 0.

IF STUCK

The record field has a name, so no positional guess is required.

Checkpoint

PROVE IT

Build a list of workshop records and identify participants uniquely.